Name: Stephen Lord

Date: March 11, 2026

Course: IT FDN 130 GP

GitHub URL: https://github.com/stephenlord10/DBFoundations-Module07/tree/main

SQL Functions

Introduction

In week seven of *IT Foundations of Database Management*, we explored views and functions. This module focused on the creation of user-defined functions (UDFs) and the logic behind applicable uses.

SQL User-Defined Function (UDF)

A SQL User-Defined Function is a function created in a relational database that performs a calculation or action, that returns a value. Albeit fun, writing repetitive queries over-and-over again can be a burden on the user. It might even create the potential for error in the development process. With a UDF, a user can write logic that will be repeated in queries and can be stored in the database itself.

As an example, in *Figure 1:1*, a UDF was created and is stored in the database (*fProductInventoriesWithPreviousMonthCountsWithKPIs*) that can be used to return results (*KPIs*) from a table (*vProductInventoriesWithPreviousMonthCountsWIthKPIs*).

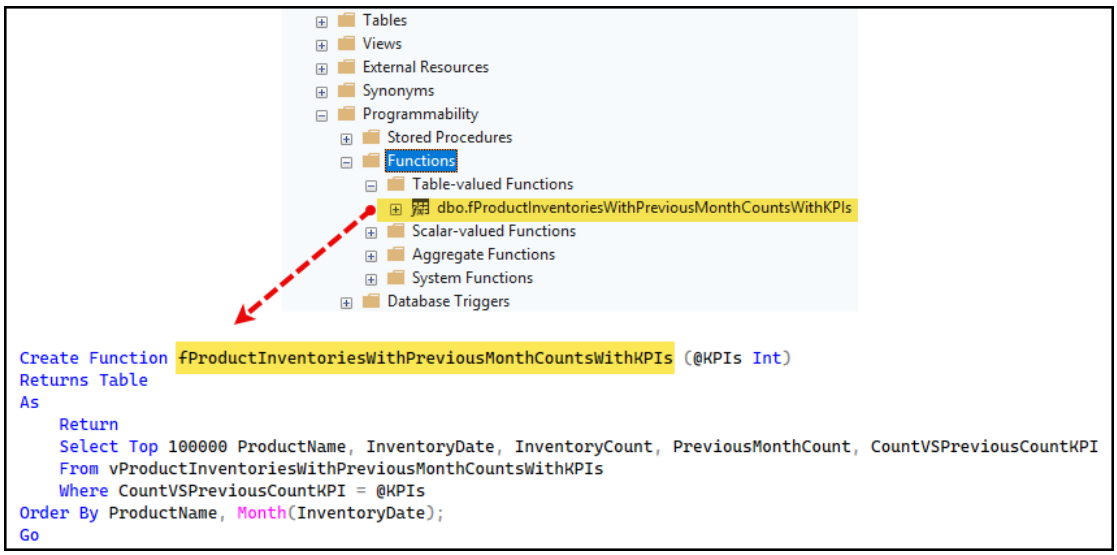

Figure 1:1 – SQL Function Stored

A user might want to use SQL UDFs for streamlined “results”, where the function has been defined (the steps) in the background and can be applied to seek the desired result. As an example, in *Figure 1:2*, the user is using the UDF (*fProductInventoriesWithPreviousMonthCountsWithKPIs*) to query KPI’s with a 1, 0, or -1. Notice how when using the *SELECT * FROM* “function” (defined value), that each individual query results in a table specific to the KPI that has a “defined value” from the script.

```
894   Select * From fProductInventoriesWithPreviousMonthCountsWithKPIs(1);
895   Select * From fProductInventoriesWithPreviousMonthCountsWithKPIs(0);
896   Select * From fProductInventoriesWithPreviousMonthCountsWithKPIs(-1);
897   Go
```

100 % 12 0

Results | Messages

	ProductName	InventoryDate	InventoryCount	PreviousMonthCount	CountVSPreviousCountKPI
1	Alice Mutton	February, 2017	10	0	1
2	Aniseed Syrup	January, 2017	13	0	1
3	Aniseed Syrup	February, 2017	23	13	1
4	Boston Crab Meat	January, 2017	123	0	1
5	Boston Crab Meat	February, 2017	133	123	1
6	Camembert Pierrot	January, 2017	19	0	1
7	Camembert Pierrot	February, 2017	29	19	1
8	Carnarvon Tigers	January, 2017	42	0	1

	ProductName	InventoryDate	InventoryCount	PreviousMonthCount	CountVSPreviousCountKPI
1	Alice Mutton	January, 2017	0	0	0
2	Alice Mutton	March, 2017	10	10	0
3	Chef Anton's Gumbo Mix	January, 2017	0	0	0
4	Chef Anton's Gumbo Mix	March, 2017	10	10	0
5	Gorgonzola Telino	January, 2017	0	0	0
6	Gorgonzola Telino	March, 2017	10	10	0
7	Perth Pasties	January, 2017	0	0	0
8	Perth Pasties	March, 2017	10	10	0

	ProductName	InventoryDate	InventoryCount	PreviousMonthCount	CountVSPreviousCountKPI
1	Aniseed Syrup	March, 2017	3	23	-1
2	Boston Crab Meat	March, 2017	113	133	-1
3	Camembert Pierrot	March, 2017	9	29	-1
4	Carnarvon Tigers	March, 2017	32	52	-1
5	Chai	March, 2017	29	49	-1

Figure 1:2 – SQL Function Queries with Result(s)

Scalar, Inline, and Multi-Statement Functions

In SQL User-Defined Functions (UDFs) there are different types of functions that return different results: scalar, inline, and multi-statement. Each one serves its own purpose.

A scalar function will return a single value as an expression. You might see them used to calculate something with the intent of yielding a single result with a quick calculation. This occurs in SQL by processing each row-by-row, then returning a single value for the entire group of data.

An inline function will return a table of results based on parameters. This function is incredibly superior in its performance, since the function itself is merged with the main query. They are simple in nature and therefore execute fast.

A multi-statement function will return a table but allows for multiple statements to be used in the logic. The downside with creating something that allows for multiple statements/logic, is that the functionality can “bog” the performance of the query. The more “logic” that goes into a multi-statement function, the more “execution” that has to occur.

Summary

Week seven of *IT Foundations of Database Management* continued expanding on the foundations of building a structurally sound database with tables and functions that can assist in streamline processes/query down the road. I found this module very useful in my own professional setting, as I see much value in the use of functions, as much of my own work is repetitive with data/finance! Looking forward to the Final.